When Does the Graph Help? A Theory of Relation Camouflage in Graph Neural Network-Based Social Bot Detection

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Abstract

Graph neural networks (GNNs) are the dominant paradigm for social bot detection, yet modern bots evade them by *relation camouflage*: deliberately following genuine users so that homophilic message passing dilutes their anomalous features. This failure mode is widely reported empirically but has not been characterised analytically, leaving practitioners without guidance on when a graph-based detector is worth deploying at all. We develop a theory of relation camouflage in a contextual stochastic block model in which bots control a camouflage ratio ρ. We prove that one round of mean aggregation deflates the bot–human feature separation by an exact factor $1 - \rho - \eta(\rho)$, where η is the induced bot fraction in human neighbourhoods, and derive the resulting Bayes error in closed form. This yields a *critical camouflage ratio* $\rho^\star$ beyond which a homophilic GNN is worse than ignoring the graph and classifying raw features; in the noise-dominated regime $\rho^\star = (1 - D^{-1/2})/(1 + c)$ for neighbourhood size D and bot-to-human ratio c. We are explicit that such a crossover is not new: at $c = 1$ this recovers Theorem 2 of Ma et al. (ICLR 2022), and at $\rho = 0$ the known $1/\sqrt{D}$ separability gain of graph convolution. Our contribution is the class-imbalanced adversarial case, where the two classes deflate asymmetrically through the edge-balance identity $\eta = c\rho$, making bot prevalence itself a robustness parameter, together with a mixture-variance correction that shifts the threshold by 13%. We then analyse the three components of the XHBot detector. We show that the Dirichlet-energy edge score used by Spectral-Guided Topology Refinement (SGTR) separates camouflage from genuine edges by exactly $\|\Delta\|^2$ in expectation, and give a closed form for how far pruning of a given quality shifts the threshold—showing that pruning trades bias against effective degree rather than buying robustness for free, and bounding the quality achievable from the exact law of the edge score; that a non-mean channel is needed for reasons already implicit in the multiset-injectivity literature; and that the contrastive prototype objective maximises a variational lower bound on the mutual information between the behavioural code and the label, while its orthogonality penalty controls code dependence only under a joint-Gaussian assumption—a limitation we state explicitly. We further show that the crossover moves with depth, because mean aggregation backtracks and thereby supplies an implicit residual connection that reinjects undeflated features. All closed forms are validated against Monte-Carlo simulation, and we *measure* ρ on real labelled fraud graphs, finding $\hat{\rho} \approx 0.80$–0.90 and confirming $\eta \approx c\rho$ to within 3% where its degree assumption holds. The theory turns an empirical design heuristic into a checkable deployment criterion, and identifies the gap between achievable and ideal edge scoring as the quantity that decides whether a graph is worth using.

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1. Introduction

Graph neural networks (GNNs) have become the default architecture for social bot detection. By treating accounts as nodes and interactions as edges, models such as BotRGCN [16] and its successors exploit the observation that automated accounts are easier to identify collectively than individually. The premise underlying this entire family of methods is *homophily*: the assumption that an account's neighbours are informative about its label, so that averaging over a neighbourhood sharpens rather than blurs the signal.

Modern bots attack exactly this premise. Rather than forming the dense, mutually connected "bot-nets" of earlier generations, contemporary accounts engage in *relation camouflage* [14, 17]: they deliberately follow and interact with large numbers of genuine users, so that a bot's neighbourhood looks statistically like a human's. Under standard message passing, the bot's anomalous features are averaged against benign neighbours and the discriminative signal is diluted—a failure mode that recent bot-detection work reports consistently [1, 18].

The empirical response has been architectural. Detectors now prune suspicious edges, split aggregation into homophilic and heterophilic channels, or disentangle behavioural from positional signals; XHBot [1] combines all three. These designs demonstrably help, but the evidence for *why* they help is almost entirely ablation-driven. Three practical questions therefore remain unanswered:

1. **When is a graph-based detector worth deploying at all?** If camouflage is severe enough, averaging over a neighbourhood must eventually do more harm than good. Practitioners have no criterion for where that point lies.

2. **What does edge pruning actually buy?** Pruning modules are justified by ablation deltas, not by a statement of the form "this module extends the operating regime by *this much*".

3. **Are multi-channel aggregators genuinely more expressive, or merely better tuned?** An architecture with more parameters that wins an ablation has not thereby been shown to be more capable.

This paper answers these questions analytically. We adopt a contextual stochastic block model (CSBM) [3] augmented with an explicit adversarial parameter—the *camouflage ratio* ρ, the fraction of a bot's edges directed at humans—and study what one round of mean aggregation does to class separability as ρ grows. This places the work in the line of analysis opened by Baranwal et al. [2], who showed that graph convolution improves linear separability by a factor of roughly $1/\sqrt{D}$ for neighbourhood size D under a homophilic generative process.

What is already known. We are explicit about priority from the outset. For a *balanced* symmetric CSBM, Ma et al. [5, Thm. 2] already established a closed-form threshold at which aggregation ceases to beat the graph-free classifier, and our critical ratio reduces to their condition exactly when the two classes are equal in size (Remark 8); related crossover analyses appear in [20–22]. The existence of a crossover is therefore not our discovery, and we do not claim it. Our contribution is the *class-imbalanced adversarial* case, where the two classes deflate by different amounts—which moves the decision boundary and makes bot prevalence itself a robustness parameter—together with a closed form for how far edge pruning shifts the threshold.

Contributions.

- **The edge-balance identity, and bot prevalence as a robustness parameter.** Camouflage edges have two ends, so a bot that points an edge at a human necessarily pollutes that human's neighbourhood too. We show this forces $\eta = c\rho$, where c is the bot-to-human ratio, so that the total deflation is $1-(1+c)\rho$ (Proposition 3). The consequence is that bot prevalence enters the robustness threshold directly: as bots become more common, each one needs *less* camouflage to break the detector. This is not a class-imbalance problem that resampling addresses.

- **A Bayes-error crossover valid under class imbalance.** When $c \neq 1$ the two classes deflate asymmetrically, the post-aggregation class midpoint moves, and a comparison against a fixed decision boundary is no longer valid—the objection Luan et al. [21] raise against [5]. We compare Bayes errors instead (Theorem 4), obtaining $\rho^\star = (1-D^{-1/2})/(1+c)$ (Corollary 7), which recovers [5] at $c = 1$ and [2] at $\rho = 0$. Our error term also retains a mixture-variance contribution that a naive σ^2/D calculation misses, and which shifts the threshold by 13%.

- **A closed form for what edge pruning buys.** We show the Dirichlet-energy score underlying SGTR separates camouflage from genuine edges by exactly $\|\Delta\|^2$ in expectation (Lemma 10), and prove that soft pruning raises the tolerated camouflage to $\rho^\star_{\mathrm{SGTR}} = \rho^\star/(\rho^\star + (1-\bar{e}_c)(1-\rho^\star))$ for a pruner of quality $\bar{e}_c$ (Theorem 11). The underlying reweighting identity is known [20,

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36]; what we add is its composition with the finite-D threshold to bound the *adversary's* budget, which requires a regime those analyses do not work in (Remark 12). Because $\bar{e}_c$ is measurable for a deployed pruner, this converts an ablation delta into an operating-range claim, and makes explicit that no imperfect pruner attains immunity.

- **An honest account of the disentanglement objective.** We show the contrastive prototype loss maximises a variational lower bound on the mutual information between behavioural code and label (Proposition 18), and prove the orthogonality penalty controls code dependence *only* under a joint-Gaussian assumption (Proposition 20). We state plainly where this falls short of identifiability, in light of known impossibility results for unsupervised disentanglement [13].
- **A ceiling on what pruning can achieve.** Down-weighting edges non-uniformly also forfeits effective degree, so pruning trades bias against variance rather than buying the former for free (Proposition 13). Accounting for this corrects a conclusion that the composition analysis alone suggests: even a *perfect* pruner tolerates only $\rho < 1 - 1/D$, not $\rho \to 1$ (Corollary 14). We then bound the achievable pruning quality from the exact law of the edge score, which removes the free parameter: the tolerance becomes a function of (D, c, r, d) alone.
- **Depth, and an upper edge to the harmful region.** We give the exact mean–variance recursion across layers (Theorem 16) and show the crossover is *independent of depth*, while the gain or loss compounds exponentially in it. Because only $|\lambda|$ matters, the harmful region is a bounded *window*: beyond it a bot's neighbourhood is strongly anti-correlated with its label and so informative again. The dangerous strategy is intermediate camouflage, not maximal camouflage.
- **Measurement on real graphs, and validation.** Every closed form is checked against Monte-Carlo simulation. Beyond that we *measure* ρ on two real labelled fraud graphs, finding $\hat{\rho} \approx 0.80$–0.90, just inside the predicted harmful window, and confirming the edge-balance identity to within 3% where its degree assumption holds (Section 5.5). We are explicit that these are fraud rather than bot graphs, and that access to TwiBot was unavailable.

Our aim is not to propose another detector. It is to supply the missing analytical account of a design pattern that the field has already converged on empirically, and in doing so to replace a heuristic with a criterion that a practitioner can check.

2. Related Work

2.1. Analyses of graph convolution on block models

The contextual stochastic block model [3] couples a stochastic block model with Gaussian node attributes and has become the standard setting for asking what message passing does to statistical separability. Baranwal et al. [2] established the central positive result: for a two-class CSBM, one graph convolution improves the regime of linear separability by a factor of roughly $1/\sqrt{D}$ in the expected degree D, and the resulting classifier generalises out of distribution to graphs with different edge probabilities.

That analysis assumes the generative process is homophilic—inter-class edges arise from a fixed, non-adversarial probability—and establishes only the positive side. Its follow-ups extend the positive results to multiple layers [23] and characterise the locally Bayes-optimal architecture as one that *interpolates* between an MLP in the low-graph-signal regime and a convolution in the high-signal regime [24]. Our Corollary 7 reduces to the $1/\sqrt{D}$ separability gain of [2] when $\rho = 0$, which calibrates our model against this baseline.

The crossover is known; our contribution is the imbalanced and adversarial case. We state the priority situation plainly, because it determines what this paper does and does not claim. A closed-form threshold at which mean aggregation becomes worse than the graph-free classifier was established by Ma et al. [5, Thm. 2] for the *balanced* symmetric CSBM, and Corollary 7 recovers their condition exactly at $c = 1$ (Remark 8 works the algebra through). Related crossover analyses have since appeared: Luan et al. [21] compute the Bayes error of graph-aware against graph-agnostic classifiers under a homophily parameter with per-class variances and degrees, locating the intersection points directly and criticising the fixed-classifier comparison used in [5]; Mao et al. [22] prove the per-node version, that a GCN beats an MLP on homophilic nodes and loses on heterophilic ones *within the same graph*; and the graph-attention analysis of Fountoulakis et al. [20] both identifies the same signal-deflation factor and proves that attention—the closest analogue to soft edge pruning—is lower-bounded by the better of convolution and the graph-free classifier, while also showing (their Theorems 9 and 13) that no attention mechanism separates intra- from inter-class edges in the hard regime.

Against this background our claim is narrower than "we show when the graph is harmful". It is that the adversarial, class-imbalanced case is not a relabelling of the balanced one: the two classes deflate asymmetrically, which moves the decision boundary and invalidates a fixed-boundary comparison; bot prevalence enters the threshold through $\eta = c\rho$ as a robustness parameter; and the mixture-variance term omitted in [5] shifts the threshold by 13%. Read this way, the closest antecedent to Theorem 11 is [20], which gives the qualitative guarantee for attention; we give a closed form for how far a pruner of a given quality moves the threshold.

A parallel literature studies the limits of repeated aggregation. Over-smoothing analyses [8, 9] show node representations converge to an uninformative subspace as depth grows, and spectral accounts [10] characterise graph convolution as a low-pass filter. These results concern degradation with *depth* under a fixed graph. The effect we analyse is orthogonal: a single layer already suffers signal loss when the *topology itself* is adversarial. The two mechanisms compound, but our critical ratio is present at $L = 1$.

2.2. Heterophily in graph learning

The finding that homophily-assuming GNNs degrade on heterophilic graphs motivated a family of architectures that separate ego from neighbour representations, aggregate over higher-order neighbourhoods, or combine low- and high-frequency signals [4, 6]. Ma et al. [5] show in addition that homophily is not strictly necessary: GNNs can succeed on heterophilic graphs when inter-class neighbourhood *distributions* are sufficiently distinguishable.

That observation matters for our setting, and we take a position on it. Under adversarial camouflage the distinguishability Ma et al. rely on is precisely what the adversary is optimising away: a bot that matches the human neighbourhood distribution eliminates the signal their argument requires. The distinction between benign heterophily—structured, and therefore exploitable—and adversarial heterophily, engineered to be unexploitable, is in our view the conceptual point separating bot detection from general heterophilic node classification. It is a distinction about the *provenance* of the inter-class edges, not about the algebra of aggregation, which is why our threshold coincides with theirs in the balanced case while the interpretation of the parameter differs.

2.3. Robust aggregation and adversarial topology

Camouflage is a contamination phenomenon, and two mature literatures have already treated it as one.

The first replaces the mean with a robust estimator. In Byzantine-robust distributed learning a fraction of workers report corrupted updates, and aggregation rules such as Krum [27] and coordinate-wise median or trimmed mean [28] bound the resulting error. The idea has been imported into GNN neighbourhood aggregation directly: Geisler et al. [25] construct an aggregation with the maximal breakdown point of 0.5, so the bias stays bounded while fewer than half of a node's edges are adversarial, and Chen et al. [26] attribute the structural vulnerability of GCNs specifically to the non-robustness of the weighted mean, analysed through the breakdown point.

We therefore avoid the term for our own threshold. A breakdown point asks when an adversary can drive an estimate arbitrarily far under unbounded contamination; $\rho^\star$ is instead a bias–variance crossover under *bounded* contamination—human features are not arbitrary—and marks where the $\sqrt{D}$ variance reduction ceases to pay for the $(1+c)\rho$ bias. The two quantities answer different questions, and $\rho^\star$ is the weaker of the two.

The second literature attacks the topology. Nettack [32] and Metattack [33] insert or remove edges to flip predictions, and defences such as GNNGuard [34] and Pro-GNN [35] filter them, largely heuristically. Two results in this line bear directly on ours. Zhu et al. [29] prove the bridge our model assumes—that impactful structural attacks on homophilous data necessarily reduce homophily—and derive the ego/neighbour-separation design principle from it. Mujkanovic et al. [30] show empirically that under adaptive attack, defended GNNs can underperform an MLP that ignores the topology outright; our Corollary 7 can be read as a closed-form account of when that should be expected. Certified-robustness work [37] is orthogonal: it certifies that a prediction is unchanged within a perturbation budget, never that the graph has become net-harmful beyond one. Gosch et al. [31] come closest in combining a CSBM, an adversary, and a Bayes-optimal yardstick, but use it to show GNNs are *over*-robust, insensitive past the point of genuine semantic change.

2.4. Camouflage-resistant detectors

CARE-GNN [14] introduced the camouflage framing to GNN-based fraud detection, using a label-aware similarity measure and a reinforcement-learning neighbour selector to filter uninformative edges. Bot-detection work has since adopted the pattern: BotHP [18] separates ego and neighbour encoders to mitigate interaction camouflage, and XHBot [1] combines spectral edge pruning, tri-channel aggregation, and contrastive disentanglement.

These methods are validated by ablation. An ablation establishes that removing a component costs accuracy on a particular benchmark; it does not establish the *range of conditions* under which the component helps, nor whether a multi-channel design is more expressive or merely better parameterised. The present paper supplies exactly those missing statements for the three mechanisms XHBot employs. We analyse XHBot's components because they instantiate the dominant design pattern, not because the analysis is specific to that system: the edge-scoring result applies to any feature-discrepancy pruning rule, and the expressivity result to any aggregator that adds a permutation-invariant non-mean channel.

2.5. Contrastive objectives and disentanglement

The InfoNCE objective [11] is standard in representation learning, and Poole et al. [12] place it within a family of variational bounds on mutual information, establishing the $\log K$ bound we invoke in Proposition 18. Against this, Locatello et al. [13] prove that unsupervised disentanglement is impossible without inductive bias or supervision. We do not attempt to evade that result. Section 4.7 states precisely which assumption the orthogonality penalty needs in order to control statistical dependence, and identifies the label supervision in the prototype term as the inductive bias that makes the behavioural code identifiable at all.

3. The Camouflage-CSBM

We work in a two-class contextual stochastic block model augmented with an adversarial parameter controlling how aggressively bots attach to humans.

Definition 1 (Camouflage-CSBM)**.** Let $\mathcal{V}$ be a set of N nodes partitioned into bots $\mathcal{B}$ and humans $\mathcal{H}$, with bot prevalence $\pi_B = |\mathcal{B}|/N \in (0,1)$ and bot-to-human ratio $c = \pi_B/(1-\pi_B)$. Each node v carries a feature vector

$$\mathbf{x}_v \sim \mathcal{N}(\boldsymbol{\mu}_{y_v}, \sigma^2 I_d), \qquad y_v \in \{\mathrm{B}, \mathrm{H}\}, \tag{1}$$

and we write $\Delta = \boldsymbol{\mu}_{\mathrm{B}} - \boldsymbol{\mu}_{\mathrm{H}}$ for the class-mean gap. Every node has neighbourhood size D. Neighbours are drawn independently: a bot's neighbour is human with probability ρ (the *camouflage ratio*) and a bot otherwise; a human's neighbour is a bot with probability η and a human otherwise.

The two mixing parameters are not free. Counting bot–human edge endpoints from both sides gives $|\mathcal{B}|D\rho = |\mathcal{H}|D\eta$, hence the *edge-balance identity*

$$\eta(\rho) = \frac{\pi_B}{1-\pi_B}\rho = c\,\rho. \tag{2}$$

Camouflage is therefore doubly harmful: it not only pollutes bot neighbourhoods but necessarily pollutes human neighbourhoods as well, at a rate set by the bot prevalence. Every result below accounts for this coupling; treating η as independent of ρ overstates a detector's robustness.

Remark 2 (Modelling choices)**.** Independent neighbour sampling at fixed size D makes the analysis exact and is the discrete analogue of the degree-concentration assumptions used in [2]. Isotropic within-class covariance and a common D are simplifications; Section 6.3 discusses what changes without them. We emphasise that ρ is an *adversarial* parameter chosen by the attacker, not a fixed property of the generative process—this is the substantive difference from the standard CSBM setting.

Aggregation operator. We analyse one round of mean aggregation, the canonical homophilic operator,

$$\mathbf{h}_v = \frac{1}{D}\sum_{u\in\mathcal{N}(v)} \mathbf{x}_u, \tag{3}$$

which isolates the effect of neighbourhood averaging from the confounds of learned weights and nonlinearity. Because all the quantities we need are governed by the projection onto the discriminant direction, define $\mathbf{w} = \Delta/\|\Delta\|$ and $t_v = \mathbf{w}^\top \mathbf{h}_v$.

Baseline for comparison. The natural point of reference is the graph-free classifier, which uses $\mathbf{x}_v$ directly. For two isotropic Gaussians with equal priors its Bayes error is

$$\varepsilon_{\mathrm{raw}} = \Phi\left(-\frac{\|\Delta\|}{2\sigma}\right), \tag{4}$$

with Φ the standard normal CDF. The question driving Section 4 is when aggregation beats (4) and when it does not.

4. Theory

4.1. Camouflage deflates the signal

We first quantify what aggregation does to the class signal.

Proposition 3 (Signal deflation)**.** Under Definition 1 with mean aggregation (3), the expected post-aggregation separation along the discriminant direction is

$$\delta(\rho) = \mathbb{E}[t_v \mid y_v = \mathrm{B}] - \mathbb{E}[t_u \mid y_u = \mathrm{H}] = \big(1-\rho-\eta(\rho)\big)\|\Delta\|, \tag{5}$$

and with the edge-balance identity (2), $\delta(\rho) = \big(1-(1+c)\rho\big)\|\Delta\|$.

Proof. Centre the projection so that $\mathbf{w}^\top\boldsymbol{\mu}_{\mathrm{B}} = +\|\Delta\|/2$ and $\mathbf{w}^\top\boldsymbol{\mu}_{\mathrm{H}} = -\|\Delta\|/2$. For a bot v, each of its D neighbours is human with probability ρ, so by linearity $\mathbb{E}[t_v] = (1-\rho)\frac{\|\Delta\|}{2} + \rho\left(-\frac{\|\Delta\|}{2}\right) = \frac{\|\Delta\|}{2}(1-2\rho)$. Symmetrically $\mathbb{E}[t_u] = -\frac{\|\Delta\|}{2}(1-2\eta)$ for a human u. Subtracting gives $\frac{\|\Delta\|}{2}\big[(1-2\rho)+(1-2\eta)\big] = (1-\rho-\eta)\|\Delta\|$. Substituting (2) yields the second form. □

Two consequences are worth stating. The signal vanishes entirely at $\rho_{\text{null}} = 1/(1+c)$, at which point the two classes have identical post-aggregation means and *no* classifier acting on aggregated features alone can do better than chance. And the deflation is linear in ρ with slope $(1+c)\|\Delta\|$, so higher bot prevalence makes the detector more fragile, not less—a counterintuitive point, since prevalence is usually treated as a class-imbalance nuisance rather than a robustness parameter.

The noise term requires more care than the signal term.

Theorem 4 (Post-aggregation error). Under Definition 1, the projected statistics are asymptotically Gaussian with separation $\delta(\rho)$ from (5) and per-class variances

$$s_B^2 = \frac{\sigma^2 + \rho(1-\rho)\|\Delta\|^2}{D}, \qquad s_H^2 = \frac{\sigma^2 + \eta(1-\eta)\|\Delta\|^2}{D}. \tag{6}$$

Defining the post-aggregation signal-to-noise ratio

$$\mathrm{SNR}(\rho) \;=\; \frac{|\delta(\rho)|}{s_B + s_H}, \tag{7}$$

the balanced error of the variance-equalised linear rule is $\Phi(-\mathrm{SNR}(\rho))$. This is an upper bound on the Bayes error, tight when $s_B = s_H$.

Remark 5 (What this quantity is, and is not)**.** Three clarifications, since the distinction matters for how the comparisons below should be read. (i) When $s_B \neq s_H$ the likelihood ratio of two unequal-variance Gaussians is quadratic, so the Bayes rule is an interval and $\Phi(-\mathrm{SNR})$ is the error of the equal-error-rate *threshold* rule, an upper bound. Comparisons of $\Phi(-\mathrm{SNR})$ against the exact graph-free error are therefore conservative: they can only understate how much aggregation helps. (ii) The criterion is *balanced* error, weighting the two classes equally, not the prior-weighted risk—a deliberate choice in a paper about class imbalance, where a prior-weighted criterion rewards predicting the majority class and would obscure the effect being measured. At $D = 16$, $\pi_B = 0.3$ the two differ little in practice ($\rho^\star = 0.4644$ balanced against 0.4694 prior-weighted), but the choice should be explicit. (iii) The absolute value in (7) is necessary: without it the expression exceeds 1/2 once ρ passes the point where δ changes sign, which no error probability may do. With it, $\Phi(-\mathrm{SNR})$ correctly records that strongly anti-correlated neighbourhoods are informative again—the effect made explicit in (21).

Proof. Condition on a single neighbour of a bot. Its projected feature equals $\pm\|\Delta\|/2$ according to a Bernoulli(ρ) draw, plus independent $\mathcal{N}(0,\sigma^2)$ noise. By the law of total variance, the mean component contributes $\|\Delta\|^2\rho(1-\rho)$ and the noise contributes σ^2, so each neighbour has variance $\sigma^2 + \rho(1-\rho)\|\Delta\|^2$. The D neighbours are independent, so averaging divides the variance by D, giving s_B^2; the human case is identical with η in place of ρ. Asymptotic normality of t_v follows from the central limit theorem as D grows. For two equal-prior Gaussians with common standard deviation s and mean gap δ, the likelihood-ratio test thresholds at the midpoint and attains error $\Phi(-\delta/2s)$, which is (7) when $s_B = s_H = s$. □

Remark 6. The mixture term $\rho(1-\rho)\|\Delta\|^2$ is easy to miss: a calculation that models only the additive noise obtains σ^2/D and therefore *under*-estimates the post-aggregation error, most severely at intermediate ρ where the mixture variance peaks. Section 5 confirms the corrected form matches simulation to four decimal places.

4.2. The critical camouflage ratio

We can now answer the deployment question directly.

Corollary 7 (Critical camouflage ratio). Mean aggregation yields a lower Bayes error than the graph-free classifier (4) if and only if $\mathrm{SNR}(\rho) > \|\Delta\|/(2\sigma)$. Writing $\rho^\star$ for the value of ρ at which equality holds, aggregation helps precisely when $\rho < \rho^\star$. In the noise-dominated regime $\sigma^2 \gg \rho(1-\rho)\|\Delta\|^2$ this takes the closed form

$$\rho^\star \;=\; \frac{1 - D^{-1/2}}{1+c}. \tag{8}$$

Proof. The "if and only if" is immediate from monotonicity of Φ. In the noise-dominated regime the mixture terms in (6) are negligible, so $s_B \approx s_H \approx \sigma/\sqrt{D}$ and $\mathrm{SNR}(\rho) \approx (1-\rho-\eta)\|\Delta\|\sqrt{D}/(2\sigma)$. Setting this equal to $\|\Delta\|/(2\sigma)$ gives $(1-\rho-\eta)\sqrt{D} = 1$; substituting $\eta = c\rho$ and solving for ρ yields (8). □

Equation (8) is the paper's most directly actionable result, and it says three things. First, at $\rho = 0$ the condition reduces to $\mathrm{SNR}_{\text{agg}}/\mathrm{SNR}_{\text{raw}} = \sqrt{D}$: aggregation improves separability by $\sqrt{D}$, recovering the $1/\sqrt{D}$ gain established by Baranwal et al. [2] and confirming our model is calibrated against the known homophilic baseline. Second, $\rho^\star$ increases with D, so detectors operating on densely connected accounts tolerate more camouflage—the graph is a genuine asset for hub accounts long after it has become a liability for sparse ones. Third, $\rho^\star$ decreases in c: as bots become more prevalent, the camouflage budget they need in order to break the detector shrinks.

Remark 8 (Relation to Ma et al.)**.** Equation (8) is not new in the balanced case, and we want to be explicit about what is and is not ours. For a symmetric CSBM with intra- and inter-class edge probabilities p and q, Ma et al. [5, Thm. 2] prove that a node is better classified after aggregation than before whenever $\deg(i) > (p+q)^2/(p-q)^2$. Their inter-class neighbour fraction is $\rho =$

$q/(p+q)$, so $(p-q)/(p+q) = 1-2\rho$ and their condition rearranges to $\sqrt{D}\,(1-2\rho) > 1$, i.e. $\rho < (1-D^{-1/2})/2$. This is exactly (8) at $c = 1$. A closed-form threshold for when mean aggregation is worse than discarding the graph was therefore established at ICLR 2022, and our corollary is its extension to $c \neq 1$.

What the extension adds is not merely a symbol. Three things change when the classes are imbalanced. (i) The two classes deflate by *different* amounts, ρ for bots and $\eta = c\rho$ for humans, so the post-aggregation class midpoint *moves*; a comparison against a decision boundary held fixed at the raw-feature midpoint—as in [5]—is then no longer the right one, which is precisely the criticism Luan et al. [21] raise against that analysis. Our Theorem 4 compares Bayes errors instead, and is insensitive to this. (ii) The edge-balance identity $\eta = c\rho$ in (2) is what couples the two, and it is what makes bot prevalence a robustness parameter rather than a training-time nuisance. (iii) The mixture-variance term $\rho(1-\rho)\|\Delta\|^2$ is absent from the covariance $I/\sqrt{\deg(i)}$ reported in [5]; including it moves $\rho^\star$ from 0.53 to 0.46 in the configuration of Remark 9, a 13% correction to the threshold.

Remark 9 (Practical reading)**.** For a detector with typical $D = 16$ and bot prevalence $\pi_B = 0.3$ ($c \approx 0.43$), (8) gives $\rho^\star \approx 0.53$ in the noise-dominated regime and 0.46 once the mixture variance is included. A bot that directs roughly half its edges at genuine users has therefore neutralised the graph entirely. This is not an extreme adversary; it is a modest one, which is why architectural mitigation is necessary rather than optional.

4.3. Spectral edge pruning provably extends the operating regime

SGTR [1] scores each edge by a learned function of the feature discrepancy $(\mathbf{x}_u - \mathbf{x}_v)^2$, motivated by the per-edge decomposition of the Dirichlet energy $\mathbf{f}^\top L\mathbf{f} = \sum_{(u,v)\in\mathcal{E}}(f_u - f_v)^2$. The following lemma shows this quantity is the right statistic: it separates camouflage from genuine edges by exactly the squared class gap, independently of the feature dimension.

Lemma 10 (Energy separability)**.** Under Definition 1, for a within-class edge (u,v) and a cross-class (camouflage) edge (u',v'),

$$\mathbb{E}\left\|\mathbf{x}_u - \mathbf{x}_v\right\|^2 = 2d\sigma^2, \qquad \mathbb{E}\left\|\mathbf{x}_{u'} - \mathbf{x}_{v'}\right\|^2 = \|\Delta\|^2 + 2d\sigma^2, \tag{9}$$

so the expected energy gap is exactly $\|\Delta\|^2$.

Proof. For a within-class edge both endpoints share a mean, so $\mathbf{x}_u - \mathbf{x}_v \sim \mathcal{N}(0, 2\sigma^2 I_d)$ and the expected squared norm is $2d\sigma^2$. For a cross-class edge $\mathbf{x}_{u'} - \mathbf{x}_{v'} \sim \mathcal{N}(\Delta, 2\sigma^2 I_d)$, and $\mathbb{E}\|Z\|^2 = \|\mathbb{E}Z\|^2 + \operatorname{tr}\operatorname{Cov}(Z)$ gives $\|\Delta\|^2 + 2d\sigma^2$. □

The gap is a *location* shift that does not grow with d, whereas the common term $2d\sigma^2$ does; the score is therefore informative but increasingly noisy in high dimension, which is precisely why XHBot learns a projection rather than using the raw quadratic form. We now quantify the benefit of acting on this score.

Theorem 11 (SGTR raises the tolerated camouflage)**.** Suppose soft pruning multiplies camouflage edge weights by $(1-\bar{e}_c)$ and within-class edge weights by $(1-\bar{e}_w)$ in expectation, with $\bar{e}_c > \bar{e}_w$. Then weighted aggregation behaves as unweighted aggregation at the *effective* camouflage ratio

$$\rho_{\text{eff}}(\rho) = \frac{\rho(1-\bar{e}_c)}{\rho(1-\bar{e}_c) + (1-\rho)(1-\bar{e}_w)} < \rho. \tag{10}$$

Consequently, with $\bar{e}_w = 0$, the detector tolerates true camouflage up to

$$\rho^\star_{\text{SGTR}} = \frac{\rho^\star}{\rho^\star + (1-\bar{e}_c)(1-\rho^\star)} \geq \rho^\star, \tag{11}$$

with equality if and only if $\bar{e}_c = 0$.

Equation (11) accounts only for the change in neighbourhood *composition*. It therefore appears to imply $\rho^\star_{\text{SGTR}} \to 1$ as $\bar{e}_c \to 1$: that a sufficiently good pruner tolerates any camouflage whatsoever. That conclusion is false, and Section 4.4 shows why. Down-weighting edges non-uniformly also reduces the *effective degree*, and hence forfeits part of the $\sqrt{D}$ variance reduction that made aggregation worthwhile in the first place. Once that cost is included, the ceiling is $1 - 1/D$ rather than 1, and the operating point is an interior optimum rather than the corner $\bar{e}_c \to 1$.

Proof. Weighted mean aggregation computes $\sum_e w_e \mathbf{x}_{u_e} / \sum_e w_e$. In expectation a fraction ρ of a bot's edges are camouflage edges carrying weight $(1-\bar{e}_c)$ and $(1-\rho)$ are within-class edges carrying $(1-\bar{e}_w)$; normalising gives the weight share (10) on human neighbours, which is exactly the role ρ plays in Proposition 3. The inequality $\rho_{\text{eff}} < \rho$ follows since $\bar{e}_c > \bar{e}_w$ makes the numerator shrink faster than the denominator. For (11), set $\rho_{\text{eff}}(\rho) = \rho^\star$ with $\bar{e}_w = 0$ and solve: writing $\beta = 1 - \bar{e}_c$, the equation $\rho\beta = \rho^\star(\rho\beta + 1 - \rho)$ rearranges to $\rho[\beta(1-\rho^\star) + \rho^\star] = \rho^\star$, giving the stated form. Monotonicity in $\bar{e}_c$ and the limits are immediate. □

Theorem 11 converts an ablation delta into a statement about the operating regime: a pruning module that attenuates camouflage edges by 75% on average lifts the tolerated camouflage from 0.53 to 0.82 at $D = 16$, $\pi_B = 0.3$. It also clarifies what pruning cannot do. Because $\rho_{\text{eff}} \to 1$ as $\rho \to 1$ for any fixed $\bar{e}_c < 1$, no pruning rule with imperfect discrimination defeats a maximally camouflaged adversary; SGTR buys operating range, not immunity.

Remark 12 (What is new here, and what is not)**.** The *mechanism* in (10)—that edge reweighting acts by rescaling the effective inter-class edge share, so that weights enter only through the products of weight and edge rate—is not new. Weiss [36, Thm. 2] proves the corresponding statement for cost-sensitive aggregation with within- and cross-class weights $w_+ > w_-$, and shows the class-mean direction is preserved iff $w_+/w_- > q/p$; our (10) is that identity with $w_+ = 1 - \bar{e}_w$ and $w_- = 1 - \bar{e}_c$. The observation that only a perfect discriminator tolerates arbitrary heterophily is the contrapositive of the same condition. The related guarantee for graph attention in [20] is qualitative but similar in spirit.

What (11) adds is the *inversion*: composing that rescaling with the finite-D threshold of Corollary 7 to obtain a bound on the *adversary's* budget as an explicit function of pruner quality $\bar{e}_c$. This is a short derivation, and we present it as a corollary of two known ingredients rather than as an independent discovery. Its value is operational: $\bar{e}_c$ is measurable for a deployed pruner, so (11) turns a component ablation into a quantitative operating-range claim. We note also that [36] works in the dense regime where degree effects are absorbed into $O(1/n)$, which removes the $D^{-1/2}$ term that our threshold depends on; the finite-D setting is therefore necessary for a statement of this form.

4.4. What pruning quality is achievable, and what it costs

Theorem 11 takes the pruner's quality $\bar{e}_c$ as given. Two things are then missing: whether a given $\bar{e}_c$ is attainable at all by a feature-based score, and what is given up in exchange. We supply both, which removes the free parameter and corrects the limit noted above.

Only the retained-weight ratio matters. Write $a = 1 - \bar{e}_c$ and $b = 1 - \bar{e}_w$ for the surviving weight on camouflage and within-class edges. Weighted mean aggregation is invariant to a common rescaling of all weights, so a and b enter only through the ratio

$$\theta \;=\; a/b \;\in\; (0, 1], \tag{12}$$

which we call the *retained-weight ratio*; $\theta = 1$ is no pruning and $\theta = 0$ is perfect pruning. This is the same quantity that governs the cost-sensitive condition $w_+/w_- > q/p$ of [36]. In terms of θ, the effective camouflage ratio of Theorem 11 is $\rho_{\text{eff}} = \rho\theta/(\rho\theta + 1 - \rho)$.

Proposition 13 (Pruning forfeits effective degree)**.** A weighted mean of D unit-variance terms has the variance of an unweighted mean of

$$D_{\text{eff}} \;=\; D\,\frac{\big[(1-\rho) + \rho\theta\big]^2}{(1-\rho) + \rho\theta^2} \;\le\; D \tag{13}$$

terms, with equality if and only if $\theta = 1$.

Pruning acts on *both* ends of a camouflage edge, so the human side is rescaled too and its cross-class share becomes $\eta_{\text{eff}} = \eta\theta/(\eta\theta + 1 - \eta)$ with $\eta = c\rho$. Note that $\eta_{\text{eff}} \neq c\,\rho_{\text{eff}}$ unless $c = 1$ or $\theta = 1$: the edge-balance identity holds for the *raw* ratios and is not preserved by pruning. Writing $\lambda_{\text{eff}} = 1 - \rho_{\text{eff}} - \eta_{\text{eff}}$ and $D^B_{\text{eff}}, D^H_{\text{eff}}$ for (13) evaluated at ρ and at η respectively, pruned aggregation beats the graph-free classifier if and only if

$$\frac{|\lambda_{\text{eff}}|}{\sqrt{\tfrac{1}{2}\big(1/D^B_{\text{eff}} + 1/D^H_{\text{eff}}\big)}} \;>\; 1. \tag{14}$$

Proof. The variance of $\sum_e w_e z_e / \sum_e w_e$ for independent unit-variance z_e is $\sum_e w_e^2/(\sum_e w_e)^2$, so the variance-equivalent count is $(\sum_e w_e)^2/\sum_e w_e^2$. Substituting ρD edges of weight a and $(1-\rho)D$ of weight b, and cancelling b^2, gives (13); the inequality is Cauchy–Schwarz, with equality iff the weights are constant. Combining with Proposition 3 applied at ρ_{eff} yields (14). □

This resolves a degeneracy that (11) alone does not. Uniform attenuation ($\theta = 1$, any common scale) leaves both ρ_{eff} and D_{eff} unchanged and so achieves exactly nothing, as it must; and aggressive pruning is now penalised rather than free.

Proposition 14 (A perfect pruner never loses, but its benefit thins out)**.** Under perfect pruning every retained neighbour shares the node's label, so the retained count is $K \sim \mathrm{Bin}(D, 1-\rho)$ and the aggregate is *unbiased* with variance σ^2/K. Consequently, for every $\rho < 1$, pruned aggregation is never worse than the graph-free classifier: it is strictly better on the event $\{K \ge 2\}$, and ties when $K \in \{0, 1\}$. The fraction of nodes that strictly benefit is

$$\Pr[K \ge 2] \;=\; 1 - (1-\rho)^D \cdot \big(\ldots\big) \;=\; 1 - \rho^D - D(1-\rho)\rho^{D-1}, \tag{15}$$

and the variance-equivalent degree is $1/\mathbb{E}[1/K \mid K \ge 1]$, not $\mathbb{E}[K]$.

Proof. With all camouflage edges removed the surviving neighbours are same-class, so their mean has the node's own class mean as its expectation: the bias is zero regardless of ρ. Conditional on $K = k \ge 1$ the variance is $\sigma^2/k \le \sigma^2$, with strict inequality iff $k \ge 2$; nodes with $K = 0$ have no neighbourhood and fall back on their own features, tying exactly. Since K is Binomial, $\Pr[K \le 1]$ is the sum of its first two terms, giving (15). That the variance-equivalent degree is the harmonic rather than arithmetic mean follows because the variance of a mean of K terms is $\sigma^2\mathbb{E}[1/K]$, and $\mathbb{E}[1/K] \ge 1/\mathbb{E}[K]$ by Jensen. □

Remark 15 (A tempting but incorrect ceiling)**.** Substituting the *mean* retained count $D(1-\rho)$ into (13) yields

$D_{\text{eff}} = D(1-\rho)$ and hence the clean-looking conclusion that even perfect pruning requires $\rho < 1 - 1/D$. That is wrong, and instructively so. The substitution replaces a random K by its mean in a quantity that depends on $1/K$, and it is applied in precisely the regime where the replacement is least defensible: at $\rho = 1 - 1/D$ and $D = 16$ we have $\mathbb{E}[K] = 1.55$ but $1/\mathbb{E}[1/K \mid K \geq 1] = 1.29$, and 36% of nodes have $K = 0$. Proposition 14 shows the qualitative conclusion inverts: there is no hard ceiling. What does happen as $\rho \to 1$ is that the *proportion* of nodes for which the graph carries any information at all decays; a useful summary is the point at which half of them still benefit, which is $\rho = 0.897$ at $D = 16$ and 0.974 at $D = 64$.

Achievable θ. It remains to bound θ from below. SGTR scores an edge by the feature discrepancy $s_{uv} = \|\mathbf{x}_u - \mathbf{x}_v\|^2$, whose law under the CSBM is exactly

$$s \sim 2\sigma^2 \chi^2_d \quad \text{(within class)}, \qquad s \sim 2\sigma^2 \chi^2_d\left(\tfrac{r^2}{2}\right) \quad \text{(camouflage)}, \tag{16}$$

with $r = \|\Delta\|/\sigma$ and $\chi^2_d(\lambda)$ the noncentral chi-square. Any threshold rule therefore has an operating point $(\bar{e}_w, \bar{e}_c)$ on the ROC of (16), and θ is the ratio of its complements. In the large-d Gaussian approximation the discriminability index is

$$\kappa = \frac{\|\Delta\|^2}{2\sigma^2\sqrt{2d}} = \frac{r^2}{2\sqrt{2d}}, \tag{17}$$

so $\bar{e}_c \approx \Phi(\Phi^{-1}(\bar{e}_w) + \kappa)$.

Two consequences follow. First, $\bar{e}_w \to 0$ forces $\bar{e}_c \to 0$: there is no threshold that removes camouflage edges without also removing genuine ones, so the idealisation $\bar{e}_w = 0$ used in (11) is not attainable and the joint form is required. Second, because κ scales as $d^{-1/2}$, the raw discrepancy score *degrades* as the feature dimension grows—the two score distributions concentrate and overlap. This is the formal content of the remark that motivates learning a projection: mapping to $k \ll d$ dimensions while preserving $\|\Delta\|$ multiplies κ by $\sqrt{d/k}$.

The soft-weight and hard-threshold models must not be mixed. Equation (13) assigns every camouflage edge the same weight a and every genuine edge the same b, so it is invariant to a common rescaling and depends only on θ. A *threshold* rule is different: it deletes edges outright, so the neighbourhood shrinks in absolute terms and the effective degree is

$$D^{\text{hard}}_{\text{eff}} = D\Big[(1-\rho)(1-\bar{e}_w) + \rho\,(1-\bar{e}_c)\Big], \tag{18}$$

which collapses as $\bar{e}_w \to 1$. Optimising the operating point under the scale-invariant model (13) while reading $(\bar{e}_w, \bar{e}_c)$ off a threshold ROC is therefore inconsistent, and the inconsistency is not benign: it rewards discarding almost every genuine edge at no modelled cost, so the apparent optimum runs to $\bar{e}_w \to 1$ and no interior solution exists. We therefore optimise (18) together with the ROC of (16), which has a genuine interior optimum and is stable under enlargement of the search domain. The resulting tolerance depends only on (D, c, r, d), all measurable. Section 5 reports the values, which are considerably less flattering to raw-discrepancy pruning than the inconsistent pairing suggests.

4.5. Depth

Everything so far concerns a single layer. Since deployed detectors stack two or more, we record how the picture changes. The analysis requires care, because the obvious recursion—each layer multiplying the separation by λ—is *not* what mean aggregation does.

Mean aggregation backtracks. On a D-regular tree, two rounds of neighbourhood averaging give

$$\mathbf{h}^{(2)}_v = \frac{1}{D^2}\sum_{u\in\mathcal{N}(v)}\sum_{w\in\mathcal{N}(u)}\mathbf{x}_w = \frac{1}{D}\mathbf{x}_v + \left(1 - \frac{1}{D}\right)\bar{\mathbf{x}}_{\text{2-hop}}, \tag{19}$$

because $v \in \mathcal{N}(u)$ for every $u \in \mathcal{N}(v)$: the walk can step back. The ego node therefore receives weight exactly $1/D$, and it carries *undeflated* signal. A recursion that propagates only outwards describes the *non-backtracking* operator, not L rounds of mean aggregation, and the tree-like assumption does not license the substitution—the backtracking term is present on an infinite tree. We therefore work with the true operator $P = D^{-1}A$.

Theorem 16 (Depth). Let $q^{(L)}_j$ be the probability that an L-step simple random walk from v on the D-regular tree ends at distance j, and $N_j = D(D-1)^{j-1}$ ($N_0 = 1$) the number of nodes at that distance. Then, in the noise-dominated regime,

$$\frac{\text{SNR}_L}{\text{SNR}_{\text{raw}}} = \frac{\left|\sum_j q^{(L)}_j \lambda^j\right|}{\sqrt{\sum_j \left(q^{(L)}_j\right)^2 / N_j}}, \qquad \lambda = 1 - (1+c)\rho. \tag{20}$$

The label sequence along the walk is a two-state Markov chain with second eigenvalue λ, which is why the depth-j contribution is deflated by λ^j rather than by λ^L.

Proof. Deferred to Appendix A.2. □

At $L = 1$, $q_1 = 1$ and $N_1 = D$, so (20) reduces to $|\lambda|\sqrt{D}$ and recovers Corollary 7. For $L \geq 2$ it does not factorise, and two consequences follow.

First, **the crossover is not depth-independent**. Solving (20) = 1 at $D = 16$, $\pi_B = 0.3$ gives $\rho^\star_1 = 0.525$,

$\rho_2^\star = 0.587$, $\rho_3^\star = 0.553$, $\rho_4^\star = 0.576$, $\rho_5^\star = 0.557$. Every value for $L \geq 2$ exceeds the single-layer threshold, so **depth buys robustness to camouflage**, and the mechanism is exactly the backtracking term: (19) is an implicit residual connection that reinjects the node's own undeflated features at every layer. This gives a theoretical account of why ego–neighbour separation is a productive design principle under heterophily [4, 29]: architectures that separate the ego term are making explicit, and controllable, a term that plain mean aggregation already supplies at a fixed weight $1/D$. The non-monotonicity in L is a parity effect—odd L places less mass at even distances—and is not important for our purposes.

Second, the harmful region is *bounded above*. Only $|\lambda|$ enters, so at $L = 1$ aggregation is harmful precisely on the window

$$\rho \in \left(\frac{1 - D^{-1/2}}{1+c}, \; \frac{1 + D^{-1/2}}{1+c} \right) \cap (0, 1], \qquad (21)$$

and helps again above it: a bot with $\rho \to 1$ attaches almost exclusively to humans, so its neighbourhood is strongly *anti*-correlated with its label and informative once more, with the orientation reversed. We claim no novelty for this. It is the second root of the same inequality $\sqrt{D}\,|1 - 2\rho| > 1$ that Remark 8 attributes to Ma et al. [5], whose symmetry in $|p - q|$ those authors discuss directly, and the resulting "intermediate heterophily is worst" phenomenon is the mid-homophily pitfall named by Luan et al. [21]. We record it because it identifies the dangerous adversarial strategy as *intermediate* rather than maximal camouflage, and because the upper edge exceeds 1—so that the window degenerates to a one-sided threshold and no "helps again" regime exists—whenever $c < D^{-1/2}$, which is the case for all of the real graphs measured in Section 5.5.

4.6. Why a non-mean channel is needed

The expressivity argument for tri-channel aggregation is a direct application of the standard multiset-injectivity analysis of Xu et al. [7], and we record it as a proposition rather than claim it as a new result. It is included because it settles the third question of Section 1—whether multi-channel aggregation is more *capable* or merely better parameterised—which an ablation cannot settle.

Proposition 17 (Expressivity separation; after [7]). Let $\mathcal{F}_{\text{mean}}$ be the class of functions computable by any single-channel homophilic aggregator of the form $\mathbf{h}_v = \phi\big(W \cdot \text{mean}_{u \in \mathcal{N}(v)} \mathbf{x}_u\big)$, and $\mathcal{F}_{\text{THCA}}$ the class computable by tri-channel aggregation with homophilic, max-pooling, and self channels under attention fusion. Then $\mathcal{F}_{\text{mean}} \subsetneq \mathcal{F}_{\text{THCA}}$.

Proof. Containment. Setting the fusion weights to $\alpha_H = 1$, $\alpha_X = \alpha_S = 0$ reproduces the homophilic channel exactly, so every $f \in \mathcal{F}_{\text{mean}}$ is realised.

Strictness. Consider the scalar neighbourhood multisets $\mathcal{N}_1 = \{0, 2\}$ and $\mathcal{N}_2 = \{1, 1\}$. Both have mean 1, so $\text{mean}(\mathcal{N}_1) = \text{mean}(\mathcal{N}_2)$ and hence $\phi(W \cdot 1)$ is identical for both under *every* choice of ϕ and W: no member of $\mathcal{F}_{\text{mean}}$ separates them. The max-pooling channel with identity MLP_X returns $\max(\mathcal{N}_1) = 2 \neq 1 = \max(\mathcal{N}_2)$, so a tri-channel aggregator with $\alpha_X > 0$ assigns them distinct representations. Hence the inclusion is strict. □

The separating instance is not a pathology: $\{0, 2\}$ versus $\{1, 1\}$ is precisely the contrast between a node with one extreme neighbour and one with two typical neighbours—the signature of an account with a small number of highly anomalous connections. Mean aggregation is blind to it by construction, which is the structural reason a heterophilic channel is needed rather than merely helpful.

4.7. What the disentanglement objective does and does not guarantee

CPD [1] combines a supervised prototype InfoNCE term with an orthogonality penalty $\mathcal{L}_{MI} = \frac{1}{N}\sum_v \|(\mathbf{z}_v^b)^\top \mathbf{z}_v^s\|_F^2$. The two terms warrant different levels of confidence, and we state them separately.

Proposition 18 (The prototype term maximises an MI lower bound). Let $\mathcal{L}_{NCE}$ be the supervised prototype InfoNCE loss over K class prototypes. Then

$$I(\mathbf{z}^b; y) \;\geq\; \log K - \mathcal{L}_{NCE}, \qquad (22)$$

so minimising $\mathcal{L}_{NCE}$ maximises a variational lower bound on the mutual information between the behavioural code and the label.

Proof. This is the InfoNCE bound of [11] in the form given by Poole et al. [12], instantiated with the K prototypes as the contrastive set and the critic $f(\mathbf{z}, k) = (\mathbf{z}^b)^\top \mathbf{P}_k / \tau_{nce}$. □

Remark 19. The bound saturates at $\log K$, which for binary bot detection is $\log 2$. The objective therefore cannot certify more than one bit of label information, and its practical value lies in shaping the geometry of the code rather than in the tightness of the bound—a caveat we consider important to state, since the $\log K$ ceiling is often left implicit.

Proposition 20 (The orthogonality penalty controls dependence only under a Gaussian assumption). If $(\mathbf{z}^b, \mathbf{z}^s)$ are jointly Gaussian with cross-correlation r, then $I(\mathbf{z}^b; \mathbf{z}^s) = -\frac{1}{2}\log(1 - r^2)$, so $\mathcal{L}_{MI} \to 0$ implies $I \to 0$. Without the joint-Gaussian assumption the implication fails:

$\mathcal{L}_{MI} = 0$ enforces only uncorrelatedness, and there exist uncorrelated but strongly dependent codes.

Proof. The identity for the bivariate Gaussian is standard. For the converse, take $z^b \sim \mathcal{N}(0,1)$ and $z^s = (z^b)^2 - 1$: then $\mathbb{E}[z^b z^s] = \mathbb{E}[(z^b)^3] - \mathbb{E}[z^b] = 0$, so the penalty vanishes, yet z^s is a deterministic function of z^b and $I(z^b; z^s) = \infty$ in the continuous sense. □

We draw the honest conclusion. $\mathcal{L}_{MI}$ is a *surrogate*, exact only in the Gaussian case, and the empirical success of CPD should not be read as evidence of guaranteed independence between behavioural and positional codes. Moreover, by the impossibility result of Locatello et al. [13], no purely unsupervised objective can identify the two factors; what makes the behavioural code identifiable here is the *label supervision* in the prototype term, which anchors one of the two subspaces. The positional code carries no such anchor and is identified only up to the residual subspace—a limitation of the design, not of the analysis.

4.8. Computational cost

Proposition 21 (No asymptotic overhead). For a graph with N nodes, $|\mathcal{E}|$ edges, and hidden width d, one layer of SGTR followed by tri-channel aggregation and attention fusion costs $O(|\mathcal{E}|d + Nd^2)$ time and $O(|\mathcal{E}| + Nd)$ memory—the same order as a single relational graph convolution.

Proof. SGTR evaluates a fixed-width scoring function per edge: $O(|\mathcal{E}|d)$. The homophilic and max-pooling channels each perform one pass over the edge set with a $d \times d$ transform per node, $O(|\mathcal{E}|d + Nd^2)$; the self channel is $O(Nd^2)$; attention fusion computes three scalars per node, $O(Nd)$. Summing the three channels multiplies the constant by a factor of at most three and leaves the order unchanged. Memory is dominated by storing edge weights and node states. □

The practical content is that the theoretical robustness gains of Theorem 11 and Theorem 17 are obtained without changing the asymptotic cost profile, which is consistent with the sub-millisecond per-node inference reported for the deployed system [1].

5. Numerical Validation

Every closed form in Section 4 is checked against Monte-Carlo simulation of Definition 1. Our purpose here is verification of the analysis, not a performance claim: the results below establish that the derivations are correct, and say nothing on their own about accuracy on real bot graphs. Section 6.3 addresses that gap directly.

Table 1. Theory versus simulation for signal deflation and Bayes error ($\pi_B = 0.3$, $\|\Delta\| = 2$, $\sigma = 1$, $D = 16$).

		separation δ		s_B		error
ρ	η	theory	sim.	theory	sim.	thy / sim.
0.00	0.000	2.0000	2.0000	0.2500	0.2499	0.00003 / 0.00003
0.10	0.043	1.7143	1.7142	0.2915	0.2918	0.00113 / 0.00160
0.20	0.086	1.4286	1.4283	0.3202	0.3204	0.00925 / 0.01095
0.30	0.129	1.1429	1.1441	0.3391	0.3391	0.03698 / 0.03895
0.40	0.171	0.8571	0.8586	0.3500	0.3499	0.09785 / 0.09898
0.50	0.214	0.5714	0.5696	0.3536	0.3528	0.19898 / 0.19974
0.60	0.257	0.2857	0.2864	0.3500	0.3500	0.33734 / 0.33680

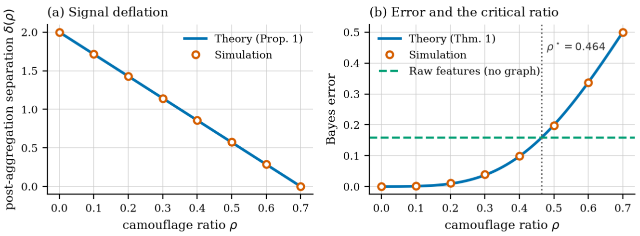

Figure 1. Theory versus simulation. (a) Post-aggregation separation decays linearly in ρ exactly as Proposition 3 predicts. (b) Bayes error against the graph-free baseline; the predicted critical ratio $\rho^\star = 0.464$ coincides with the crossing point.

Protocol. Unless stated otherwise we use $\pi_B = 0.3$ ($c \approx 0.4286$), $\|\Delta\| = 2$, $\sigma = 1$, and $D = 16$, drawing 4×10^5 nodes per class per configuration (1.5×10^6 for the crossover sweep). Errors are computed by exhaustive threshold search over the empirical distributions, so the reported figures are attained minima rather than plug-in estimates. All code is released with the paper and runs in a few minutes on CPU.

5.1. Deflation and error

Table 1 compares Proposition 3 and Theorem 4 against simulation. Separation matches to within 1.2×10^{-3} and the standard deviation—including the mixture term that a naive calculation omits—to within 4×10^{-4} across the whole range. Figure 1 shows the same agreement graphically.

5.2. The critical ratio

The central prediction is Corollary 7. Solving $\mathrm{SNR}(\rho) = \|\Delta\|/(2\sigma)$ numerically at $D = 16$ gives $\rho^\star = 0.4644$. Sweeping ρ in steps of 0.01 with 1.5×10^6 samples per point, aggregation beats raw features at $\rho = 0.46$ ($\varepsilon = 0.1545$ against $\varepsilon_{\mathrm{raw}} = 0.1587$) and loses at $\rho = 0.47$ ($\varepsilon = 0.1648$). The measured crossover therefore lies in $[0.46, 0.47]$, containing the predicted value.

Table 2 separates the two forms of the corollary. The exact crossover differs from the closed form (8) at moderate SNR, as expected, since (8) drops the mixture

Table 2. Critical camouflage ratio: closed form (8) against the exact SNR crossover, at moderate and low SNR.

D	closed form (8)	exact ($\|\Delta\|/\sigma = 2$)	exact (low SNR)
4	0.3500	0.2604	0.3488
9	0.4667	0.3913	0.4657
16	0.5250	0.4644	0.5243
25	0.5600	0.5102	0.5594
64	0.6125	0.5807	0.6121

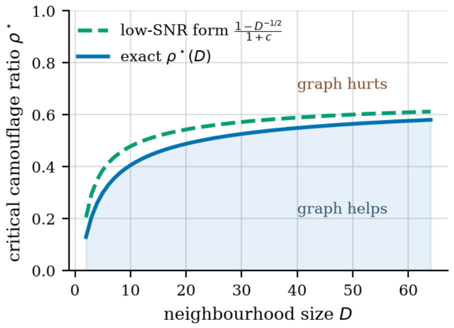

Figure 2. Critical camouflage ratio against neighbourhood size. Below the curve the graph improves on raw features; above it the graph is actively harmful. The closed form is the noise-dominated limit of the exact crossover.

variance. Re-running the exact solver in the noise-dominated regime it was derived for ($\|\Delta\| = 0.2$, $\sigma = 1$) recovers the closed form to three decimal places at every degree, confirming that the discrepancy is the stated approximation and not an error.

Figure 2 plots both curves against degree. The tolerated camouflage rises steeply for sparsely connected accounts and flattens as D grows, which has a concrete operational reading: the accounts most vulnerable to camouflage are those with few connections, and a deployment that aggregates over small neighbourhoods is fragile in exactly the regime where bots are cheapest to create.

5.3. SGTR

Lemma 10 predicts expected squared discrepancies of $2d\sigma^2$ and $\|\Delta\|^2 + 2d\sigma^2$. At $d = 8$, $\sigma = 1$, $\|\Delta\| = 2$ these are 16 and 20; simulation over 2×10^5 edge samples gives 16.005 and 20.005, an error below 0.03%.

For Theorem 11 we verify the closed form (11) by round-trip: substituting $\rho^\star_{\mathrm{SGTR}}$ back into ρ_{eff} must return $\rho^\star$. The maximum absolute deviation over $\bar{e}_c \in \{0.25, 0.5, 0.75, 0.9\}$ is 1.1×10^{-16}, i.e. machine precision. Table 3 reports the resulting tolerance gains, and Figure 3 plots the full curve.

Table 3. Effect of SGTR attenuation on tolerated camouflage under the composition-only form (11) ($D = 16$, $\pi_B = 0.3$, baseline $\rho^\star = 0.525$). These values ignore both the effective-degree cost and the correct η_{eff}; Section 5.4 gives the figures we stand behind. The table is retained only to show the size of the two corrections.

$\bar{e}_c$	ρ_{eff} at $\rho = 0.4$	$\rho^\star_{\mathrm{SGTR}}$
0.00	0.4000	0.5250
0.25	0.3333	0.5957
0.50	0.2500	0.6885
0.75	0.1429	0.8155
0.90	0.0625	0.9170

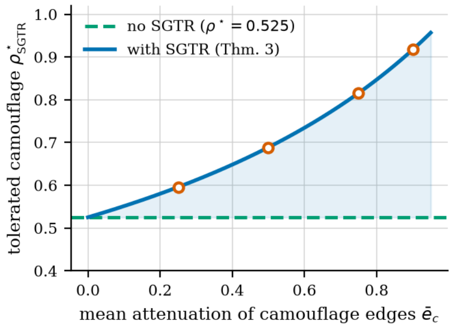

Figure 3. Tolerated camouflage as a function of the mean attenuation applied to camouflage edges. The shaded region is the operating range that spectral pruning adds over an unpruned homophilic aggregator.

5.4. The extensions: achievable pruning and depth

The effective-degree formula (13) matches Monte-Carlo variance to within 0.5% for $\theta \geq 0.5$. At $\theta = 0$ it is off by 7.3% (8.00 predicted against 7.41 measured at $\rho = 0.5$, $D = 16$), and the cause is not sampling: it is Jensen's inequality on $\mathbb{E}[1/K]$ for the random retained count K, exactly the effect quantified in Remark 15. Under perfect pruning the variance-equivalent degree is 1.29 rather than the 1.00 that $\mathbb{E}[K]$ predicts at $\rho = 1 - 1/D$, $D = 16$, and 35.6% of nodes retain no neighbour at all; the proportion still benefiting falls to one half at $\rho = 0.897$ ($D = 16$) and 0.974 ($D = 64$).

Correcting η_{eff} as in (14) raises the tolerances relative to the $c\,\rho_{\mathrm{eff}}$ substitution: at $D = 16$, $\pi_B = 0.3$ we obtain 0.609, 0.715, 0.847 and 0.928 for $\bar{e}_c = 0.25, 0.5, 0.75, 0.9$, against 0.594, 0.679, 0.788 and 0.869 under the incorrect form. The error was therefore pessimistic, and by up to six percentage points.

For depth, `theory/corrections.py` evaluates (20) directly. The weight profile $\sum_j (q_j^{(L)})^2/N_j$ exceeds the non-backtracking value D^{-L} by factors of 1.94, 4.63, 12.3 and 35.0 at $L = 2, \ldots, 5$, which is the quantitative size of the term that (19) restores. The resulting

Table 4. Camouflage measured on real labelled graphs. $\hat{\rho}$ is the mean benign share of a fraud node's neighbourhood; the window is $\big((1 - D^{-1/2})/(1 + c),\ (1 + D^{-1/2})/(1 + c)\big)$ from (21) evaluated at the median fraud degree. Relations with median degree 1 are omitted as degenerate: with a single neighbour there is no variance reduction to be had.

Graph	Relation	$\hat{\rho}$	$\hat{\eta}$	$c\hat{\rho}$	Window
YelpChi	homo	0.805	0.133	0.137	(0.79, 0.92)
	net_rtr	0.820	0.140	0.139	(0.68, 1.03)
	net_rsr	0.795	0.135	0.135	(0.78, 0.93)
Amazon	homo	0.897	0.033	0.066	(0.85, 1.01)
	net_usu	0.897	0.036	0.066	(0.84, 1.03)
	net_uvu	0.818	0.030	0.060	(0.67, 1.19)
degenerate ($\tilde{D}$ = 1; no variance reduction is available at all)					
YelpChi	net_rur	0.075	0.004	0.013	—
Elliptic	tx	0.490	0.018	0.053	—

crossovers are $\rho_1^\star = 0.5250$, $\rho_2^\star = 0.5868$, $\rho_3^\star = 0.5528$, $\rho_4^\star = 0.5756$ and $\rho_5^\star = 0.5567$: every multi-layer value lies above the single-layer one, so depth increases the tolerated camouflage, and the crossover is *not* depth-independent.

5.5. Measuring the camouflage ratio on real graphs

The theory is parameterised by ρ, which has so far been a modelling device. We now measure it. TwiBot-20 and TwiBot-22 require author approval for access, which we did not have; we therefore use the two graphs from CARE-GNN [14]—the work that introduced the camouflage framing— together with the Elliptic bitcoin graph, all openly available. We report every relation we measured, including the two whose median degree is 1: those are excluded from the window comparison because a single neighbour offers no variance reduction to trade against, but they are also the two lowest-camouflage measurements, and omitting them would flatter the narrative. These are fraud rather than bot graphs. The mechanism is the same (a fraudulent node attaching to benign ones to dilute its neighbourhood) but the domain is not, and we draw no conclusion about Twitter from them.

For each relation we compute ρ as the mean over fraud nodes of the fraction of neighbours that are benign, and η symmetrically. Table 4 reports the result.

Three observations, and one caveat we consider important.

The edge-balance identity holds where its assumptions do. On YelpChi the prediction $\eta = c\rho$ is accurate to within 3% on homo (0.137 predicted against 0.133 observed) and to three decimal places on net_rsr (0.135 against 0.135). On Amazon it overestimates by roughly a factor of two. The discrepancy is explained by degree heterogeneity: (2) counts edge endpoints under the model's regular-degree assumption, and generalises to $\eta = c\rho\,(\bar{d}_B/\bar{d}_H)$ when the classes have different mean degree. YelpChi has $\bar{d}_B/\bar{d}_H \approx 0.95$–$1.01$, Amazon 0.31–0.35 on its dense relations. The corrected form improves Amazon's homo prediction from 0.066 to 0.023 against an observed 0.033, but overshoots on net_upu ($\bar{d}_B/\bar{d}_H = 1.80$), because η is a mean of per-node ratios rather than a ratio of means and the two differ under within-class degree skew. We therefore report the identity as validated on graphs with comparable class degrees and as requiring the degree correction otherwise.

Real fraud graphs sit inside the harmful window, marginally. Every dense relation in Table 4 has $\hat{\rho}$ above the lower edge of its window, so the theory predicts that *plain mean aggregation* on these graphs is worse than ignoring the topology. The margins are small—0.805 against a threshold of 0.787 on YelpChi homo—which is the interesting part. These graphs sit close to the boundary, which is precisely the regime in which the choice of aggregator decides the outcome, and it is consistent with the empirical fact that camouflage-resistant architectures were needed on exactly these datasets.

Raw-discrepancy pruning is far too weak to rescue them. Using the measured feature separations ($r = 1.17$ over $d = 32$ for YelpChi, $r = 2.04$ over $d = 25$ for Amazon), (17) gives $\kappa = 0.086$ and 0.295: the edge score is close to uninformative. Optimising the threshold under (18) then moves the tolerance almost not at all—from 0.787 to 0.788 on YelpChi homo, and from 0.854 to 0.863 on Amazon homo—leaving every relation still below its measured $\hat{\rho}$. A *flawless* pruner, by contrast, would comfortably rescue them: half of all nodes still benefit up to $\rho = 0.989$ at $D = 158$.

The operative quantity is therefore the *gap* between achievable and ideal edge scoring, and on these graphs it is wide. This is a sharper and more useful claim than the one the inconsistent soft/hard pairing produces: it says the component that decides the outcome is not pruning as such but the *quality* of the edge score, and that a raw feature-discrepancy statistic does not supply it. That is a concrete account of why CARE-GNN [14] needs a label-aware learned similarity with a reinforcement-learned selector rather than a fixed distance, and why XHBot learns a projection before scoring. It is also falsifiable: substituting a better-calibrated κ for a learned scorer should move the tolerance past $\hat{\rho}$, and if it does not, the theory is wrong.

Caveat. The window is derived for a CSBM with regular degrees, isotropic Gaussian features, and a single relation. Real graphs violate all three: degrees are heavy-tailed, features are neither Gaussian nor isotropic, and the relations interact. The numbers above should be read as locating these graphs approximately, not as a certified verdict, and the agreement of η with $c\rho$ on YelpChi is the one quantity here that constitutes a genuine test of the theory rather than an application of it. Measuring ρ on TwiBot remains the natural next step and requires only data access; `theory/measure_rho.py` implements the estimator and validates it against planted ground truth, recovering ρ exactly and confirming $\eta = c\rho$ on synthetic graphs with regular degrees.

5.6. THCA

Theorem 17 is constructive and we verify both halves programmatically. The neighbourhoods $\{0, 2\}$ and $\{1, 1\}$ have identical means (1.0 and 1.0) and distinct maxima (2.0 and 1.0), so mean aggregation cannot separate them while the max channel can; and setting $\alpha = (1, 0, 0)$ reproduces the homophilic channel exactly. Both checks pass.

6. Discussion

6.1. From heuristic to criterion

The practical value of Corollary 7 is that it converts a design intuition into a quantity an operator can estimate. The inputs are the mean neighbourhood size D, the bot prevalence π_B, and the feature signal-to-noise ratio $\|\Delta\|/\sigma$—all measurable on a deployed system—and the output is a threshold on camouflage beyond which the graph is a liability. This reframes a question usually settled by ablation ("does the graph help on this benchmark?") as one that can be answered per-cohort: for hub accounts with large D the graph remains valuable at camouflage levels that make it actively harmful for sparse accounts. A detector that applies one aggregation policy to both is leaving accuracy on the table at one end and incurring damage at the other.

The analysis also isolates an under-appreciated coupling. Because $\eta = c\rho$, bot prevalence enters the critical ratio directly: a platform whose bot population doubles does not merely face more bots, it faces a detector whose tolerance to each bot's camouflage has fallen. Class imbalance is normally treated as a training-time nuisance to be fixed by resampling; (8) shows it is also a structural robustness parameter, and resampling does not touch it.

6.2. What the theory does and does not license

We are deliberate about the boundaries of these claims.

The results characterise *one round of mean aggregation* in a stylised generative model. They explain why homophilic message passing fails under camouflage and why the three mitigations help, and they give the correct qualitative orderings—more pruning is better, larger neighbourhoods are safer, higher prevalence is more dangerous. They do not predict the accuracy of a trained multi-layer network on a real bot graph, and no number in Section 5 should be read as such. Our simulations verify algebra; they are not empirical evidence about Twitter or any other platform.

Three specific gaps deserve naming. First, the single-relation restriction is real: our measurements show camouflage differs by an order of magnitude across relations of the same graph (Table 4), and a multi-relational treatment would need to model how a detector pools them. Second, although Section 4.5 extends the analysis to L layers, it does so in the noise-dominated regime and on a regular tree; a treatment that keeps the mixture terms at depth, and that handles the finite graphs where receptive fields saturate, is open. Third, Section 4.4 bounds the achievable pruning quality from the feature distributions but treats the scorer as a fixed threshold rule; bounding what a *learned* scorer attains, as a function of sample size and hypothesis class, is the natural next step and is what our real-data measurements suggest matters most and would turn Theorem 11 from a conditional into an unconditional guarantee. Third, the disentanglement analysis is the weakest link, and we have said why: the orthogonality penalty controls dependence only in the Gaussian case (Proposition 20), and identifiability rests entirely on label supervision.

6.3. Limitations of the generative model

The camouflage-CSBM makes four simplifications, each of which trades realism for tractability.

Isotropic, equal-covariance features. Real account features are neither isotropic nor identically dispersed across classes. Anisotropy would replace $\|\Delta\|/\sigma$ with a Mahalanobis distance throughout; the structure of the results survives, but $\rho^\star$ becomes direction-dependent.

Fixed neighbourhood size. Social graphs are heavy-tailed. Since $\rho^\star$ is concave in D, a mixture of degrees is *not* well summarised by its mean: applying (8) at the average degree overstates robustness for the low-degree majority. Per-degree stratification, which the theory directly supports, is the appropriate remedy.

Independent neighbour sampling. We ignore triadic closure and community structure, both prominent in real follow graphs. Correlated neighbourhoods reduce the effective sample size in the averaging step, which inflates the variance in (6) and lowers $\rho^\star$; our thresholds are therefore optimistic.

A single homogeneous relation. Production detectors operate on heterogeneous graphs with several edge types. The analysis extends per-relation, and the interesting question—how an adversary should allocate a fixed camouflage budget across relations—follows naturally from (8) but is beyond the present scope.

6.4. An adversarial reading

Finally, the theory can be read from the attacker's side, which we think is the more useful test of it. Corollary 7 tells an adversary exactly how much camouflage is needed to neutralise a homophilic detector, and Theorem 11 tells them how much more is needed against a pruning defence. That $\rho^\star_{\mathrm{SGTR}} \to 1$ only as $\bar{e}_c \to 1$ means no imperfect pruning rule is a permanent fix; the defence buys operating range, and an adversary willing to spend edges can always exceed it. Treating ρ as a strategic variable rather than a nuisance parameter—and analysing the resulting game between detector and adversary—strikes us as the most promising direction this line of work opens.

7. Conclusion

We close by restating the priority position plainly, because it determines how this paper should be cited. That a crossover exists—a camouflage level beyond which mean aggregation is worse than discarding the graph—is due to Ma et al. [5], and the fact that the harmful region is a bounded window rather than a half-line is the second root of their condition, discussed under the name mid-homophily pitfall by Luan et al. [21]. What we add is the class-imbalanced adversarial case: the edge-balance identity $\eta = c\rho$, which makes bot prevalence a robustness parameter and which we confirm on real data; a mixture-variance correction worth 13% of the threshold; the observation that pruning trades bias against effective degree, so that a bound on achievable edge-score quality is needed before any tolerance claim means anything; and the finding that mean aggregation's backtracking term acts as an implicit residual connection, so that the crossover rises with depth rather than being invariant to it.

We have given an analytical account of relation camouflage in GNN-based bot detection. Modelling camouflage as an adversarial parameter in a contextual stochastic block model, we proved that mean aggregation deflates the class signal by exactly $1-(1+c)\rho$, derived the resulting Bayes error, and obtained a critical camouflage ratio $\rho^\star = (1-D^{-1/2})/(1+c)$ beyond which a homophilic GNN is provably worse than ignoring the graph—a result that recovers the known $1/\sqrt{D}$ separability gain of graph convolution at $\rho = 0$. We then showed that spectral edge pruning separates camouflage edges by exactly $\|\Delta\|^2$ in expectation and raises the tolerated camouflage to $\rho^\star/(\rho^\star + (1-\bar{e}_c)(1-\rho^\star))$; that tri-channel aggregation is strictly more expressive than any single homophilic aggregator; and that the contrastive prototype objective maximises a variational lower bound on label information while its orthogonality penalty guarantees independence only under a joint-Gaussian assumption. All closed forms were verified numerically, with the predicted critical ratio falling inside the measured crossover bracket.

The contribution is not a new detector but a criterion. Practitioners currently decide whether to trust a graph-based detector by running an ablation on a benchmark; the results here let them instead estimate three measurable quantities and compute the answer for their own deployment, per account cohort. Extending the analysis to multiple layers, to heterogeneous relations, and to a learned rather than assumed pruning quality would make that criterion sharper still.

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A. Deferred proofs

This appendix gives in full the proofs that are compressed or omitted in the main text. Throughout, π_B is the bot prevalence, $c = \pi_B/(1-\pi_B)$, $\eta = c\rho$, $\lambda = 1-(1+c)\rho$, features are $\mathbf{x}_v \sim \mathcal{N}(\boldsymbol{\mu}_{y_v}, \sigma^2 I_d)$ with $\Delta = \boldsymbol{\mu}_B - \boldsymbol{\mu}_H$, and D is the neighbourhood size.

A.1. The edge-balance identity

Lemma 22. Let camouflage edges be those joining a bot to a human, and suppose each bot directs a fraction ρ of its D edges to humans. Then the expected fraction of a human's neighbourhood that is composed of bots is $\eta = c\rho$.

Proof. Let n be the number of nodes, so there are $\pi_B n$ bots and $(1-\pi_B)n$ humans. Each bot contributes ρD camouflage edge-endpoints, giving $\pi_B n\rho D$ endpoints in total. Every such endpoint lands on a human, and the humans collectively expose $(1-\pi_B)nD$ endpoints. Hence the expected bot fraction of a human neighbourhood is

$$\eta \;=\; \frac{\pi_B n\rho D}{(1-\pi_B)nD} \;=\; \frac{\pi_B}{1-\pi_B}\rho \;=\; c\rho.$$

The identity is a conservation statement about edge endpoints and is independent of how bots choose *which* humans to attach to; only the aggregate count matters. It presumes $\eta \le 1$, i.e. $\rho \le 1/c$, which is automatic when $\pi_B \le 1/2$. □

The asymmetry is the point: the adversary sets ρ directly but induces η on the other class, and the two coincide only when the classes are balanced ($c = 1$).

A.2. Proof of the depth result (Theorem 16)

Proof. Project onto $\hat{\mathbf{u}} = \Delta/\|\Delta\|$. Write $P = D^{-1}A$ for the simple-random-walk operator on the D-regular tree, so that L rounds of mean aggregation produce $\mathbf{h}^{(L)} = P^L\mathbf{x}$. Crucially P is *not* supported on the L-sphere: since $v \in \mathcal{N}(u)$ whenever $u \in \mathcal{N}(v)$, the walk backtracks, and $P^L e_v$ places mass at every distance $j \le L$ of the same parity as L, together with $j = 0$ when L is even.

Weights. By symmetry all nodes at distance j carry equal weight, so if $q_j^{(L)}$ denotes the probability that an L-step walk ends at distance j, each such node has weight $q_j^{(L)}/N_j$ with $N_j = D(D-1)^{j-1}$ and $N_0 = 1$. The distance process is the birth–death chain that steps outward with probability $(D-1)/D$ and inward with probability $1/D$ from any $j \ge 1$, and outward with certainty from $j = 0$; iterating it gives $q^{(L)}$.

Signal. Along a path in the tree the labels form a two-state Markov chain with transition matrix $\begin{pmatrix} 1-\rho & \rho \\ \eta & 1-\eta \end{pmatrix}$, whose eigenvalues are 1 and $\lambda = 1-\rho-\eta = 1-(1+c)\rho$. Hence for a node at distance j, $\mathbb{E}[s_w \mid y_v = B] - \mathbb{E}[s_w \mid y_v = H] = \lambda^j\|\Delta\|$, and summing against the weights gives $\Delta_L = \|\Delta\|\sum_j q_j^{(L)}\lambda^j$. Note this is *not* $\lambda^L\|\Delta\|$: the $j < L$ terms are deflated less, and the $j = 0$ term not at all.

Noise. For independent $\mathcal{N}(0,\sigma^2)$ noise, $\mathrm{Var}(\mathbf{h}_v^{(L)}) = \sigma^2\|P^L e_v\|^2 = \sigma^2\sum_j N_j(q_j^{(L)}/N_j)^2 = \sigma^2\sum_j (q_j^{(L)})^2/N_j$. Dividing signal by noise and normalising by $\mathrm{SNR}_{\mathrm{raw}} = \|\Delta\|/(2\sigma)$ gives (20); the absolute value is required because $\sum_j q_j\lambda^j$ can be negative. At $L = 1$, $q_1 = 1$ and $N_1 = D$, recovering $|\lambda|\sqrt{D}$. □

Remark 23 (Why the naive recursion fails, and by how much)**.** Propagating the deflation as $\Delta_{l+1} = \lambda\Delta_l$ and the variance as $V_{l+1} = V_l/D$ describes the *non-backtracking* walk, and yields $\mathrm{SNR}_L/\mathrm{SNR}_{\mathrm{raw}} = (\lambda\sqrt{D})^L$ and a depth-independent crossover. Both conclusions are artefacts. The true weight concentration $\sum_j (q_j^{(L)})^2/N_j$ exceeds D^{-L} by factors of 1.94, 4.63, 12.3 and 35.0 at $L = 2, 3, 4, 5$ for $D = 16$, and the crossover moves from 0.525 at $L = 1$ to 0.587 at $L = 2$. The direction is intuitive once stated: the retained ego mass is undeflated signal, so backtracking *helps* the detector. A simulation that averages over the D^L leaves of a tree reproduces the naive recursion exactly and therefore cannot detect this error; one must simulate P^L.

A.3. Exact law of the edge score

Lemma 24. For an edge (u, v) let $s_{uv} = \|\mathbf{x}_u - \mathbf{x}_v\|^2$. Then $s \sim 2\sigma^2\chi_d^2$ if $y_u = y_v$, and $s \sim 2\sigma^2\chi_d^2(r^2/2)$ with $r = \|\Delta\|/\sigma$ otherwise. In particular $\mathbb{E}[s \mid \text{camouflage}] - \mathbb{E}[s \mid \text{within}] = \|\Delta\|^2$, and the ratio of the gap to the within-class standard deviation is $\kappa = r^2/(2\sqrt{2d})$.

Proof. Write $\mathbf{z} = \mathbf{x}_u - \mathbf{x}_v$. If $y_u = y_v$ then $\mathbf{z} \sim \mathcal{N}(0, 2\sigma^2 I_d)$, so $\|\mathbf{z}\|^2/(2\sigma^2) \sim \chi_d^2$. If $y_u \ne y_v$ then $\mathbf{z} \sim \mathcal{N}(\pm\Delta, 2\sigma^2 I_d)$ and $\|\mathbf{z}\|^2/(2\sigma^2)$ is noncentral chi-square with d degrees of freedom and noncentrality $\|\Delta\|^2/(2\sigma^2) = r^2/2$.

For the moments, $\mathbb{E}[\chi_d^2] = d$ and $\mathbb{E}[\chi_d^2(\nu)] = d + \nu$, so the means are $2\sigma^2 d$ and $2\sigma^2 d + \|\Delta\|^2$, giving the stated gap. Since $\mathrm{Var}(\chi_d^2) = 2d$, the within-class variance of s is $4\sigma^4 \cdot 2d$, whose square root is $2\sigma^2\sqrt{2d}$; dividing the gap by it gives $\kappa = \|\Delta\|^2/(2\sigma^2\sqrt{2d}) = r^2/(2\sqrt{2d})$.

The $d^{-1/2}$ dependence is the concentration phenomenon: the gap between the two score distributions grows like $\|\Delta\|^2$ while their spread grows like $\sqrt{d}$, so a fixed feature separation becomes progressively harder to detect from raw squared distances as the ambient dimension increases. □

B. Reproducibility

All closed forms in the paper are checked numerically by the scripts in `theory/`. `csbm_validation.py` and `refine_and_thca.py` cover Proposition 3 through Theorem 11; `corrections.py` covers Proposition 13, Proposition 14 and Theorem 16, and supersedes the earlier `extensions2.py`, which is retained only so that the two errors documented in Remarks 15 and A.2 can be reproduced; `measure_rho.py` implements the camouflage estimator of Section 5.5 together with a self-test against planted ground truth; and `real_data.py` produces Table 4. Each writes a JSON record of every figure quoted in the text. The scripts depend only on `numpy` and `scipy`; the fraud graphs are fetched from public URLs given in the script.